

Efficiency and durability of tin halide perovskite photovoltaics

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Abstract

Tin halide perovskites (THPs) have emerged as promising lead-free alternatives for perovskite photovoltaics, achieving power conversion efficiencies (PCEs) approaching 18% while avoiding the use of toxic lead. In this Review, we discuss techniques for increasing the performance of THPs towards that of lead-based devices. The precursor and redox chemistry governing defect formation and oxidation processes in THPs influences the crystallization quality of THP films during solution processing. These bottlenecks include rapid crystallization kinetics and intrinsic susceptibility of Sn^{2+} to oxidation, limiting their photovoltaic performance and device lifespans. Improved device performance can be achieved through material and device engineering strategies such as crystallization regulation, intrinsic and extrinsic antioxidation, mixed-dimensional heterostructures and interfacial engineering. Further challenges to device efficiency, durability and scalable manufacturing are discussed, including routes to enabling ambient fabrication. Commercialization and large-scale fabrication of efficient and durable THPs could provide a thin-film photovoltaic technology that mitigates the toxicity limitations of more technologically mature lead-based perovskite materials.

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Key points

- Tin halide perovskite (THP) materials have similar optoelectronic and processability characteristics to their lead-based analogues, making them suitable for thin-film photovoltaics, and THPs avoid the use of toxic lead.
- Rapid crystallization and easy oxidation of THPs result in thin-film defect densities of 10^{18} cm^{-3} , which are two orders of magnitude higher than those of lead-based perovskites, limiting their performance and durability.
- The benchmark power conversion efficiency (PCE) of 17.71% for THP solar cells only reaches half of the theoretical maximum, indicating the potential for efficiency gains by suppressing defect-induced performance losses.
- Crystallization regulation through precursor solution engineering and interfacial engineering using self-assembled monolayers reduce defect-induced efficiency losses and prolong device lifespans.
- THP solar cells can achieve device lifetimes of 1,550 h at maximum power point tracking under one-sun illumination in nitrogen.
- Future commercialization of THP solar cells will require further PCE improvement, long-term operational stability and scalable manufacturing in ambient conditions.

Introduction

Sustained advances in compositional engineering, fabrication processes and device architectures have propelled the power conversion efficiency (PCE) of single-junction perovskite solar cells to 28.0% by 2026, whereas perovskite–silicon and perovskite–perovskite tandem photovoltaics (PVs) have reached certified PCEs of 35.5% and 30.1%, respectively¹. These PCE breakthroughs are primarily achieved using lead-based perovskites as light absorbers. However, the potential leakage of biohazardous Pb^{2+} ions into aquatic and terrestrial ecosystems raises concerns for human health and environmental safety², potentially impeding the large-scale deployment of perovskite solar cells for future utility-scale and residential PV applications.

In the pursuit of lead-free alternatives, tin halide perovskites (THPs) have emerged as promising absorber materials owing to their low toxicity, ideal narrow bandgaps ($E_g = 1.3\text{--}1.4 \text{ eV}$) and optoelectronic properties comparable to those of their lead-based analogues^{3,4}. The first tin halide perovskite solar cells (THPSCs) were independently reported in 2014, delivering initial PCEs of 5.7% and 6.4%, respectively^{5,6}. Despite encapsulation and dark storage in nitrogen, these prototype devices exhibited rapid efficiency decay within 24 h, along with limited device reproducibility. This instability primarily originates from the easy oxidation of Sn^{2+} to Sn^{4+} owing to its intrinsically active $5s^2$ electrons. Meanwhile, the high-energy $5p$ orbital of Sn^{2+} leads to Lewis acidic SnX_2 ($\text{X} = \text{I}, \text{Br}$ or Cl), rendering fast reactions with organic halides and, thus, accelerated crystallization of highly defective THP films⁷. As a result, abundant trap states in the bulk and at the surface of THP films, together with Sn^{4+} impurities, induce severe non-radiative recombination, markedly deteriorating their PV performance and device lifespans.

Crystallization regulation, antioxidation and interfacial modification of THPSCs have been used to address these performance and stability challenges. These have been achieved through methods including stable intermediate adducts^{8,9}, 2D templated growth^{10,11}, interfacial passivation^{12,13} and novel charge transport layers^{14,15}. For instance, a caesium-containing triple-cation composition was developed to regulate colloidal chemistry and crystallization kinetics, achieving homogeneous 2D–3D perovskite heterostructures (PHSs) that enabled a certified PCE of 16.65% with stable operation over 1,500 h in early 2025 (ref. 16). Cyanovinyl phosphonic acid-based self-assembled monolayers (SAMs) were employed as uniform interlayers to modify the buried interface, delivering a certified PCE of 17.71% and stable operation beyond 1,550 h (ref. 17). Despite promising advances in compositional, additive and interfacial engineering, the benchmark efficiency of THPSCs has only reached ~50% of their theoretical maximum of 33.4%¹⁸, highlighting inferior film quality and spontaneous oxidation. Previous reviews have focused on material innovations^{19–21}, fundamental properties^{22,23}, crystallization dynamics^{7,24–26} and degradation mechanisms^{27–29}. Enabling efficient, durable and eco-friendly PV technologies requires developing more innovative device engineering strategies based on a timely assessment of the current status, critical knowledge gaps and technological frontiers of THPSCs.

In this Review, we introduce the mechanistic fundamentals of defect and redox chemistry of THPs. After tracing the developmental trajectory of THPSCs during 2014–2026, we outline the key challenges that limit the efficiency and stability of THPSCs. Then, we present an overview that compares emerging device engineering strategies, spanning crystallization control, antioxidation, dimensional and interfacial engineering, and scalable fabrication. Finally, remaining technical obstacles and future opportunities are examined to provide a roadmap towards commercially viable THPSCs.

Defect and redox chemistry

Metal halide perovskites typically adopt a general chemical formula of ABX_3 , where A is a monovalent cation (methylammonium, MA^+ ; formamidinium, FA^+ ; caesium, Cs^+), B is a divalent metal cation (Pb^{2+} , Sn^{2+}) and X are halides (I^- , Br^- , Cl^-) (Fig. 1a). Despite the similar ionic radii of Pb^{2+} (119 pm) and Sn^{2+} (118 pm), Sn^{2+} with an electronic configuration of $[\text{Kr}]4d^{10}5s^2$ has a metastable divalent state due to its relatively weak nuclear attraction to the outer $5s^2$ electron pair (Fig. 1b). In contrast, the $6s^2$ electron pair of Pb^{2+} ($[\text{Xe}]4f^{14}5d^{10}6s^2$) is stabilized by strong nuclear attraction, resulting from inadequate electron shielding of the $4f$ orbital. The chemically active $5s^2$ electron pair of Sn^{2+} facilitates its spontaneous oxidation to Sn^{4+} and confer high Lewis acidity to SnI_2 , facilitating its reaction with formamidinium iodide (FAI) to accelerate the crystallization of THPs. The excessively fast crystallization kinetics and undesirable oxidation lead to highly defective THP films with defect densities of $\sim 10^{18} \text{ cm}^{-3}$, two orders of magnitude higher than the $10^{15}\text{--}10^{16} \text{ cm}^{-3}$ of their lead-based counterparts³⁰.

Defect chemistry

Subjected to uncontrolled crystallization and mild post annealing ($\sim 80^\circ \text{C}$), these THP films possess point defects in bulk crystals and extended defects at the surface and grain boundaries. Taking CsSnI_3 as an example, the tin vacancy (V_{Sn}) has the lowest formation energy among 12 possible native point defects, followed by the caesium vacancy (V_{Cs}), iodine interstitial (I_i) and Cs–Sn anti-site (Cs_{Sn}) under tin-rich conditions³¹. The formation energy of V_{Sn} becomes negative under iodine-rich and moderate growth conditions, suggesting the easy

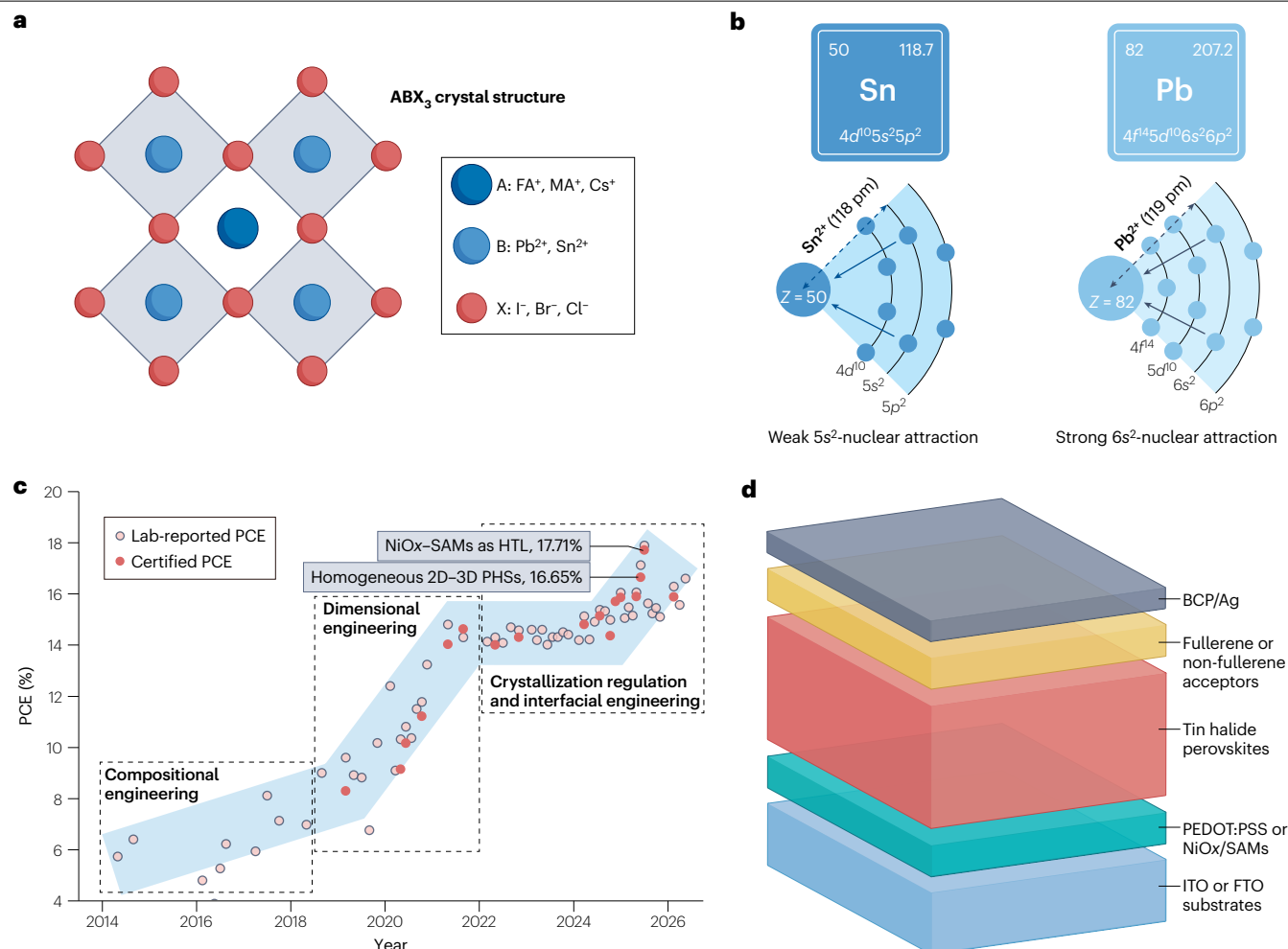


Fig. 1 | Performance evolution and device structure. **a**, ABX₃ crystal structure. **b**, Electronic configuration of tin and lead elements. **c**, Power conversion efficiency (PCE) evolution of tin halide perovskite solar cells (THPSCs) from 2014 to 2026 (data available in Supplementary Table 1). **d**, Typical p-i-n device architecture of THPSCs, where poly(3,4-ethylenedioxythiophene) polystyrene

sulfonate (PEDOT:PSS) or bilayer nickel oxide (NiO_x) and self-assembled monolayers (SAMs) are employed as the p-type hole transport layers (HTLs), and fullerene derivatives or non-fullerene acceptors are used as the n-type electron transport layers (ETLs). Cs⁺, caesium; FA⁺, formamidinium; MA⁺, methylammonium; PHS, perovskite heterostructure.

formation of large numbers of V_{Sn} defects in CsSnI₃ films regardless of growing conditions. Substituting small inorganic Cs⁺ cations (188 pm) with larger organic cations of MA⁺ (217 pm) and FA⁺ (253 pm) weakens the antibonding coupling between Sn-5s and I-5p and elongates the Sn-I bond, thereby increasing the formation energy of V_{Sn} for reduced background hole densities³². Hence, compared with unipolar p-type CsSnI₃ and MASnI₃, FASnI₃ exhibits more intrinsic conductivity under tin-rich conditions, rendering it a suitable absorber for PV applications. In this regard, adding excessive tin precursors or SnF₂ additives in the precursor solutions can create tin-rich conditions to suppress V_{Sn} formation and thus improve the crystalline quality of THP films. The cation effect is further evidenced by the delayed hot-carrier cooling (2.8 ns) in FASnI₃ compared with that of CsSnI₃ (0.6 ns), which can be attributed to a reduced background V_{Sn} density induced by compressive lattice strain³³.

Non-equilibrium crystallization of THP films can arise from incomplete precursor conversion, solvent residues or phase segregation.

These unoptimized processing conditions develop macroscopic imperfections such as cracks, pinholes, dislocations, precipitates and impurities³⁴. These structural inhomogeneities originate from disrupted nucleation-growth dynamics and manifest as extended defects at surfaces and grain boundaries, inducing the formation of undercoordinated Sn²⁺ sites, dangling I⁻ bonds and oxidized Sn⁴⁺ species that impair charge transport and initiate oxidation. For instance, inadequate colloidal stabilization in precursor solutions can yield sparse intermediate-phase nuclei, limiting grain coalescence during Ostwald ripening and promoting pinhole formation in THP films³⁵. Moreover, owing to the intrinsically rapid crystallization kinetics of THPs, delayed annealing of wet perovskite films might trap residual solvent within the crystalline matrix, generating voids at the buried interface upon ultimate solvent evaporation³⁶. These extended defects induce substantial PV performance losses and accelerate the oxidation of THP films from the surface to the bulk crystals upon external stimuli. To mitigate extended defects, precursor solution engineering

Box 1 | Efficiency progress of tin halide perovskite photovoltaics

Phase I. Device structure and compositional engineering (2014–2019)

Starting in 2014, prototype tin halide perovskite solar cells (THPSCs) adopted a conventional n–i–p device architecture, delivering power conversion efficiencies (PCEs) of around 6% and limited device stability of less than 24 h (refs. 5,6). The inverted p–i–n structure solar cells then became prevalent because of their non-oxidizing charge transport layers and, thus, high device reproducibility¹²⁴. Meanwhile, compositional engineering, particularly A-site regulation, also contributed to the PCE increase of THPSCs. Compared with archetypal methylammonium-based tin halide perovskites (THPs), mixed-cation compositions offer more controllable crystallization kinetics and improved oxidation resistance during film formation, leading to compact and dense THP layers and PCEs exceeding 8%^{90,125–127}.

Phase II. Dimensional engineering and crystallization regulation (2020–2022)

The conceptual design of 2D–3D perovskite heterostructures (PHSs) enabled improvements to both the efficiency and stability of THPSCs. Alternative 2D spacer cations have been incorporated as precursor additives or post-treatment agents to guide the oriented growth of 3D THPs, enabling 2D–3D PHSs with controlled crystallization and improved robustness^{10,11,95}. In parallel, precursor solution engineering emerged as an effective method to regulate crystallization kinetics. Intermediate adduct complexes were developed to facilitate homogeneous nucleation and delayed growth of high-quality THPs.

The resulting films had markedly fewer pinholes, uniform grain orientations and carrier diffusion lengths up to 290 nm (refs. 9,64). Integrating these two advances suppresses defect formation during solution processes, and thus enables the PCEs of THPSCs to increase beyond 14%.

Phase III. Colloidal chemistry and interfacial engineering (2023–2025)

Colloidal chemistry advances and interfacial engineering further propelled the PCE increase of THPSCs. Typically, a pyridyl-based fullerene derivative of TPPF was developed as a coordinative binder to regulate crystallization⁶⁷. The resulting THP films exhibited improved crystallinity and interfacial band alignment, yielding a high short-circuit current density (J_{sc}) of 24.81 mA cm^{−2} and a certified PCE of 15.14%. Inserting a dipolar interlayer at the top surface further reduced interfacial photovoltage losses, leading to a high open-circuit voltage (V_{oc}) of 0.97 V and a certified PCE of 15.7%¹². Subsequently, Cs⁺ cations were introduced to promote 2D and 3D colloidal aggregation in the precursor solution, synchronizing the nucleation kinetics of 2D and 3D THPs for the development of homogeneous 2D–3D PHSs¹⁶. These phase segregation-free THP layers exhibited improved crystallinity and preferential orientations, delivering a certified PCE of 16.65%. In parallel, self-assembled monolayer (SAM) interlayers were introduced at the buried interface between THP and nickel oxide (NiO_x) layers, guiding the growth of high-quality THP films¹⁷. These SAM-modified THPSCs demonstrated a certified PCE of 17.71% with operational stability of 1,550 h.

strategies such as stable intermediate adducts, anti-solvent-assisted crystallization and 2D perovskite templates have been widely adopted to fabricate dense, compact and fully covered THP films.

Oxidation mechanisms

With a low redox potential of Sn⁴⁺/Sn²⁺ (+0.15 V versus standard hydrogen electrode)³⁷, THPs are highly susceptible to oxidation, driven by both intrinsic and extrinsic factors. Although Sn⁴⁺ impurities in the precursor solutions can be effectively mitigated by adding tin powder or other oxygen scavengers³⁸, the oxidizing atmosphere induced by evaporated dimethyl sulfoxide (DMSO) solvent and trace amounts of oxygen present inside the glovebox during fabrication can readily promote Sn²⁺ oxidation during THP film fabrication³⁹. Under dark storage, neutral oxygen from the ambient environment can quickly degrade ASnI₃ to SnI₄, which subsequently reacts with H₂O and O₂ to release reducing iodine (I₂) species⁴⁰. Given a larger redox potential of I₂/I[−] (+0.54 V) than Sn⁴⁺/Sn²⁺ (+0.15 V), these volatile I₂ species can further oxidize ASnI₃ to generate more SnI₄, accelerating the cyclic degradation of THPs.

Intrinsically, owing to a positive oxidation potential of Sn⁰/Sn²⁺ (+0.13 V in acidic medium), Sn²⁺ undergoes spontaneous disproportionation into Sn⁴⁺ and metallic tin, which subsequently generates additional V_{sn} and releases two holes to the valence band, and thus induces p-type doping of THPs²⁷. This intrinsic Sn²⁺ oxidation can be exacerbated by light, where photo-induced superoxide (O₂[−]) species can deprotonate organic FA⁺ cations and cleave Sn–I bonds, triggering subsequent I₂ volatilization for the irreversible decomposition

of THPs²⁸. Although neutral oxygen ingress can be effectively mitigated by physical encapsulation²⁹, achieving long-term operational stability requires THP crystals durable against chemical, thermal and photochemical stresses. This intrinsic robustness requires precise crystallization regulation to optimize film microstructure and suppress V_{sn} formation in the bulk, together with holistic surface passivation to mitigate undercoordinated Sn²⁺ species at the top surface, grain boundaries and buried interface²⁰.

Efficiency progress and challenges

With sustained progress in material and device engineering during 2014–2026, the state-of-the-art THPSCs have achieved benchmark PCEs approaching 18%, with stable operation exceeding 1,550 h (refs. 16,17) (Fig. 1c and Box 1). Although promising PCEs have been demonstrated on small-area devices with an inverted p–i–n structure under controlled environments (Fig. 1d), their viability for large-scale manufacturing under ambient conditions remains unvalidated. This persistent engineering gap between laboratory-scale demonstrations and commercially viable production leaves THPSCs at a relatively low technology readiness level of 4.

According to the Shockley–Queisser (SQ) limit, the state-of-the-art THPSCs based on FASnI₃ with a bandgap of 1.4 eV exhibit substantial PV performance losses, including more than 20% photocurrent density losses (>6 mA cm^{−2})^{41,42} and a large open-circuit voltage (V_{oc}) deficit (>100 mV)⁴³ relative to their achievable values for J_{sc} of 32.88 mA cm^{−2}, V_{SQ} of 1.14 V and fill factor FF_{SQ} of 0.89 (Fig. 2a and Supplementary Table 1). The large short-circuit current density (J_{sc}) losses originate

from trap-assisted non-radiative recombination, insufficient light harvesting and limited charge-carrier diffusion lengths, whereas V_{oc} losses are mainly associated with interfacial energy-level misalignment and inadequate surface passivation⁴⁴. The abundant trap states in the bulk film and at the surface promote Sn^{2+} oxidation under dark storage and facilitate iodide migration to exacerbate cyclic degradation of THP films under light and thermal stresses, therefore undermining both shelf and operational stability of THPSCs (Fig. 2b and Supplementary Table 2). This section discusses the key bottlenecks that limit their PV performance, long-term operational stability and processability towards practical applications.

Defects and surface oxidation

Despite effective crystallization control using intermediate adducts and 2D perovskite templates, these THP films achieving benchmark PCEs still retain high defect densities (Fig. 2c). Pinholes and surface imperfections were routinely observed in their surface and cross-sectional scanning electron microscopy images, whereas pronounced trap states in the bulk have been probed by synergetic characterizations, including short average charge-carrier lifetimes (20–30 ns), high trap densities

(10^{15} – 10^{16} cm^{-3}), large Urbach energy (50–100 meV) and limited charge diffusion lengths (400–500 nm)^{16,45}. Trap-induced non-radiative recombination impairs charge transport for attenuating photocurrent, while also accelerating Sn^{2+} oxidation and initiating THP decomposition during operation³⁰ (Fig. 2d).

Although raw material purification and colloidal chemistry regulation of precursor solutions can effectively suppress Sn^{4+} formation in the bulk during THP crystallization, undesirable Sn^{4+} species (1–5 mol%) remain detectable at the film surface^{16,46}. These surface Sn^{4+} species originate from spontaneous oxidation during post annealing and subsequent layer deposition, amplified by gradient diffusion of neutral oxygen or oxidative solvents from the fabrication and ageing atmosphere⁴⁷. Intrinsically, undercoordinated Sn^{2+} sites at the top surface have lower activation energies than those within the bulk perovskite film, rendering the surface defects more susceptible to oxidation and self-disproportionation. This susceptibility can generate surface-enriched Sn^{4+} species under external stimuli such as oxygen, moisture and oxidative chemical vapour. These surface Sn^{4+} species are prone to propagate into the bulk by dissociating the weak Sn–I bonds, triggering chain decomposition of THP films^{42,48}.

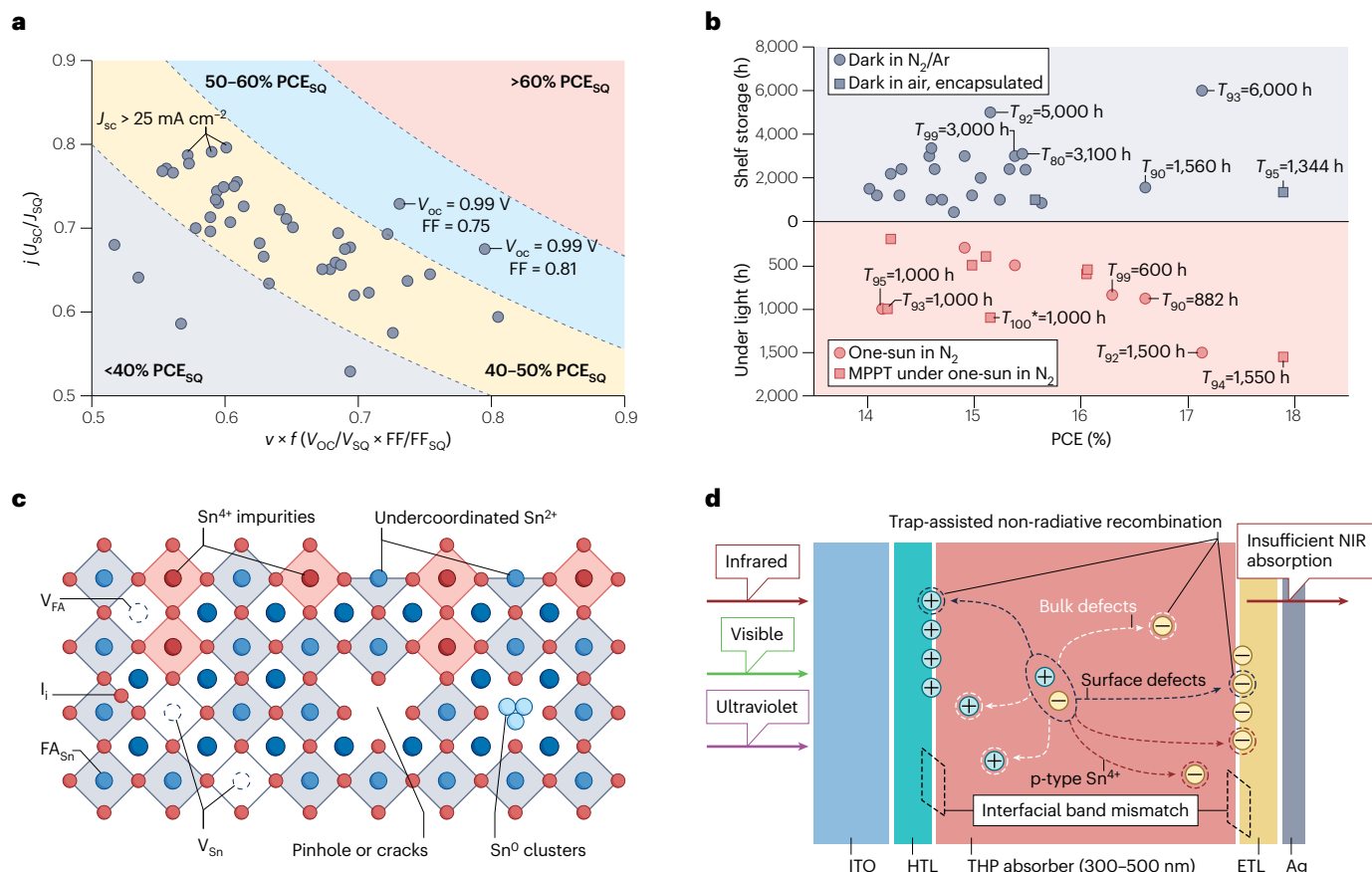


Fig. 2 | Limiting factors for efficiency and durability. **a**, Photovoltaic (PV) parameter comparison between reported tin halide perovskite solar cells (THPSCs) and Shockley–Queisser (SQ) limit. j , v and f are the ratios of the measured short-circuit current density (J_{sc}), open-circuit voltage (V_{oc}) and fill factor (FF) to their SQ limit values of $J_{sc,SQ}$, $V_{oc,SQ}$ and FF_{SQ} (data available in Supplementary Table 1). **b**, Long-term stability of THPSCs under dark storage and light illumination (data available in Supplementary Table 2). T_x refers to the

time that retains more than $x\%$ of the initial power conversion efficiency (PCE). **c**, Primary point defects and extended defects in tin halide perovskite (THP) films. **d**, Major PV performance losses in THPSCs, including trap-assisted non-radiative recombination losses from bulk defects, surface defects and p-type Sn^{4+} impurities, along with interfacial band mismatch and insufficient near-infrared light (NIR) absorption. ETL, electron transport layer; HTL, hole transport layer; I_i , iodine interstitial; V_{Sn} , tin vacancy.

Device structure and upscaling fabrication

THPs generally exhibit lower optical absorption coefficients (10^4 – 10^5 cm $^{-1}$) than their lead-based counterparts ($\sim 10^5$ cm $^{-1}$), requiring thicker films (>600 nm) for sufficient near-infrared (NIR) light harvesting^{49–51}. A thicker THP light absorber, however, demands more crystallization control to reduce bulk defect densities, whereas limited charge-carrier diffusion lengths inhibit efficient charge transport from the bulk film to the interfaces. Such a thickness–quality trade-off compromises the external quantum efficiency response in the range of 700–900 nm, leading to parasitic absorption losses and pronounced photocurrent losses^{16,17}. Most THPSCs directly adopt device architectures derived from their lead-based counterparts, typically using poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) and fullerenes as the hole transport layer (HTL) and electron transport layer (ETL), respectively. However, compared with their lead-based counterparts, the upshifted conduction band and valence band edges of THPs render these conventional transport materials energetically suboptimal, leading to energy-level mismatches that cause interfacial photovoltage losses.

Although alternative transport layers, such as indene-C₆₀ bis-adduct (ICBA) and SAM-modified NiO_x, have enabled a benchmark V_{oc} of THPSCs approaching 1.0 V (refs. 14,17), further optimization of ETLs and HTLs remains crucial, given the commonly reported V_{oc} deficits of more than 150 mV (ref. 43). Considering the hygroscopic nature of PEDOT:PSS and the high oxidizing activity of NiO_x, inserting a bridging hole transport interlayer becomes essential to prevent chemical degradation at the buried interface and extend the device lifespan for practical applications^{15,52}. Meanwhile, the synthesis costs of fullerene-based ETLs are relatively high compared with those of conventional small molecules and polymers⁵³, necessitating the development of low-cost non-fullerene electron acceptors for more affordable THPSCs.

Although small-area THPSCs with PCEs above 17% have been demonstrated on the laboratory scale, their large-area fabrication under ambient conditions remains challenging, owing to rapid crystallization and easy oxidation of THPs. To improve the technology readiness level and reduce manufacturing costs of THPSCs, scalable solution-processing techniques, such as slot-die coating and blade coating, for preparing high-quality THP films need to be validated under at least dry-air conditions. Despite initial attempts of depositing large-area THP films (10–20 cm 2) by blade coating and slot-die coating, these printable THPSCs only delivered PCEs below 10%, highlighting substantial PV performance losses from laboratory-scale demonstration to module-level upscaling^{54–56}.

Emerging device engineering strategies

Device engineering strategies, including crystallization regulation, intrinsic and extrinsic antioxidation, 2D–3D heterostructures and interfacial engineering, have been developed to improve film quality and suppress non-radiative recombination in the bulk film and at the interfaces, thereby minimizing PV performance losses and extending device lifetimes under operational conditions towards practical applications. New ETLs and HTLs have been judiciously selected to modify interfacial band alignment, and the scalable printing of THP mini-modules under a controlled atmosphere has been demonstrated. Built on these technical advances, promising device engineering strategies of THPSCs are highlighted (Supplementary Table 3).

Crystallization regulation

According to the LaMer model, THP crystallization generally proceeds via three stages, including colloidal stabilization, nucleation

and growth⁷. The stability of THP colloids in the precursor solution is mainly determined by the total repulsive potential (W_{total}) in the dispersing medium by integrating the van der Waals attraction (W_a) and electrical repulsion (W_r) between two THP colloidal particles. Such colloidal dynamics in the precursor solution determine the size and density of nuclei, whereas solute consumption during nucleation governs subsequent growth kinetics.

Solvent engineering. In typical one-step deposition of THP films, anti-solvent treatment quickly extracts DMF and DMSO solvents to reach supersaturation, inducing abrupt nucleation by substantial solute consumption within 1–2 s (ref. 57). This narrow nucleation window enables the precipitation of dense and uniform nuclei, followed by prolonged growth to produce high-quality THP crystals. Analogous to single-crystal growth, pre-nucleated seeds combined with slow growth rates yields large crystals with a minimal defect density. Accordingly, high-quality THP films can be achieved by stabilizing colloidal clusters as premature crystalline templates in the precursor solution, followed by fast nucleation and slow growth²⁴ (Fig. 3a).

Given the high Lewis acidity of Sn²⁺, stable colloidal clusters can readily form through chemical interactions between octahedral [SnX₆]^{4–} and electron-donating solvent molecules. For instance, the characteristic S=O group of DMSO can donate a lone electron pair to Sn²⁺, inducing chemical coordination of SnI₂·(DMSO)_x adduct complexes in the precursor solution⁹. These stable intermediate phases act as pre-nucleated seeds to lower the nucleation energy barrier, enabling accelerated nucleation followed by retarded crystal growth for low defect-density THP films (Fig. 3b). The Lewis basicity of solvents can be quantitatively described by their Gutmann donor numbers (DNs), with DMF and DMSO exhibiting DN of 26.6 kcal mol $^{-1}$ and 29.8 kcal mol $^{-1}$, respectively^{58,59}. Introducing high-DN solvents, such as *N,N*-diethylformamide (DN of 30.9 kcal mol $^{-1}$), *N,N*-dimethylpropylene urea (DN of 33.0 kcal mol $^{-1}$) and hexamethylphosphoramide (DN of 38.8 kcal mol $^{-1}$), can strengthen precursor–solvent coordination and promote complete dissolution of SnI₂ species, inducing the formation of monodispersed adducts for homogeneous nucleation of THPs⁶⁰. Unlike DMSO, which gradually oxidizes Sn²⁺, these neutral solvents can offer a protective chemical environment to suppress Sn⁴⁺ formation during the crystallization of THP films. With higher dielectric constants, ionic liquids such as methylammonium acetate, formamidinium acetate and butylammonium acetate can further strengthen adduct coordination and promote colloidal coagulation during precursor ageing^{61–63}. The resulting enlarged colloidal clusters lead to fewer but larger nuclei, facilitating the growth of large-grain THP films with reduced defect densities.

Additive engineering and two-step deposition. Beyond solvent engineering, intermediate adducts can be further tailored by introducing Lewis base additives containing oxygen, nitrogen, sulfur or phosphorus donors, which can interact with iodostannate species ([SnX₆]^{m–}) to form stable colloidal clusters and accelerate nucleation. For instance, urea-based additives such as trimethyl urea and 2-(2-aminoethyl)-2-thiopseudo urea can induce Lewis acid–base coordination between C=O and Sn²⁺ and the concurrent hydrogen bond of N–H and I[–] species^{64–66} (Fig. 3c). These synergistic interactions promote the formation of large and uniform colloidal clusters in the precursor solution, enabling accelerated nucleation and prolonged growth to produce smooth, compact THP films with suppressed trap states⁶⁶. The coagulation dynamics of THP colloids can be further regulated by additive–precursor coordination strength and surface zeta potentials

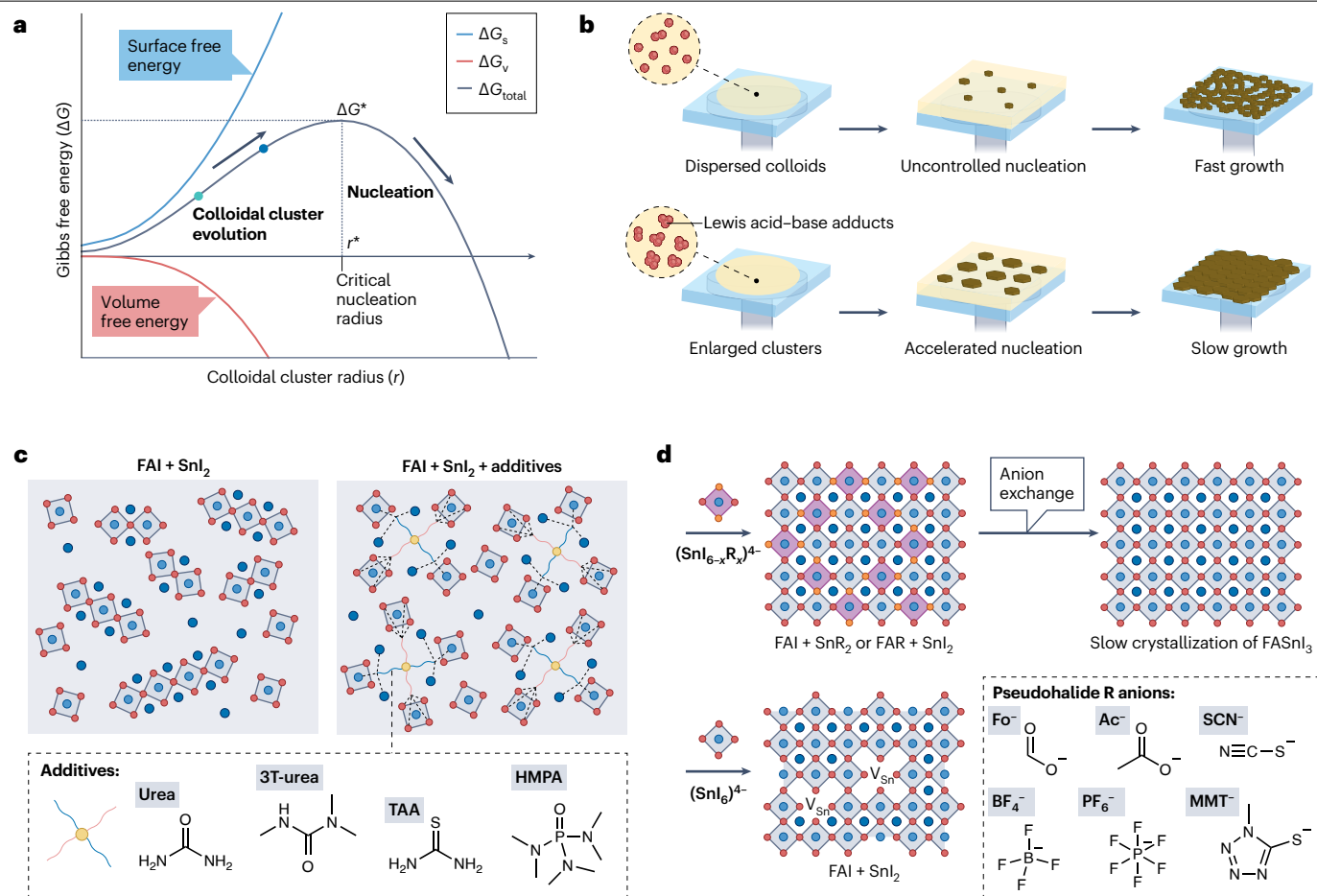


Fig. 3 | Crystal regulation and additive engineering. **a**, Gibbs free energy evolution during the stages of colloidal cluster stabilization and nucleation. According to Classic Nucleation Theory, the overall excess Gibbs free energy (ΔG_{total}) between a small solid particle of solute with a spherical radius r is equal to the sum of the surface free energy (ΔG_s) and the volume free energy (ΔG_v). Along with colloidal size evolution, nucleation occurs when ΔG_{total} reaches its critical point (ΔG^*). **b**, Chemical coordination between precursors and

solvents or additives leads to Lewis acid–base adducts, which serve as stable intermediate phases to accelerate nucleation and retard fast growth to achieve high-quality tin halide perovskite (THP) films. **c**, Colloidal dynamic variations with and without donor additives. **d**, Indirect two-step crystallization of THPs mediated by pseudohalide anions. Ac^- , acetate; BF_4^- , tetrafluoroborate; FAI, formamidinium iodide; FO^- , formate; MMT^- , 5-mercapto-1-methyltetrazole; PF_6^- , hexafluorophosphate; SCN^- , thiocyanate; V_{Sn} , tin vacancy.

at their electric double-layer configuration. Dynamic light scattering can non-invasively probe colloidal size distributions and surface electrical potential variations, revealing time-dependent colloidal evolution in the THP precursor solutions³⁵. For instance, fullerene derivatives featuring spatially isomeric pyridine terminals can bridge adjacent THP colloids to increase the average cluster size from 400 nm to 660 nm (ref. 67), whereas introducing multivalent 3-aminopyrrolidine (APD^{2+}) cations into the stern layer of THP colloids can markedly reduce the surface zeta potential from -4.22 mV to 0.69 mV (ref. 68). Either increased coordination sites or reduced surface electrostatic repulsion promotes colloidal aggregation, leading to enlarged clusters in the precursor solution and, thus, accelerated nucleation for achieving high-quality THP films.

To slow down the rapid crystallization of THPs, indirect crystallization strategies, such as non-halide precursors and 2D to 3D phase transition pathways, have been developed. For instance, formamidinium acetate (FA source) combined with NH_4I (iodine source) can replace

the commonly used FAI precursor⁶⁹, whereas SnI_2 can be substituted by $Sn(SCN)_2$ or $Sn(MMT)_2$ (MMT = 5-mercapto-1-methyltetrazole)^{70–72}. These pseudohalide precursors prolong reaction pathways and impede the fast crystallization of THPs through anion-exchange mediation (Fig. 3d). Compared with one-pot crystallization, two-step sequential deposition of SnI_2 and FAI was less frequently explored for THPs, as rapid interfacial reactions impede subsequent FAI diffusion and retard complete SnI_2 conversion into $FASnI_3$ (ref. 73). To overcome this limitation, 2D phase templates can be pre-embedded within the amorphous SnI_2 layer by adding 2D spacer cations in the first-step precursor solution⁷⁴. Subsequent FAI treatment induces a 2D to 3D phase transition via cation exchange, enabling quasi-epitaxial growth of high-quality THP films. Although THPSCs fabricated by two-step methods only delivered PCEs of 14.03% (ref. 74), these protocols enable much delayed crystallization with a wider processing window, offering more opportunities for delicate crystallization control and thus achieving quasi-single-crystalline THP films.

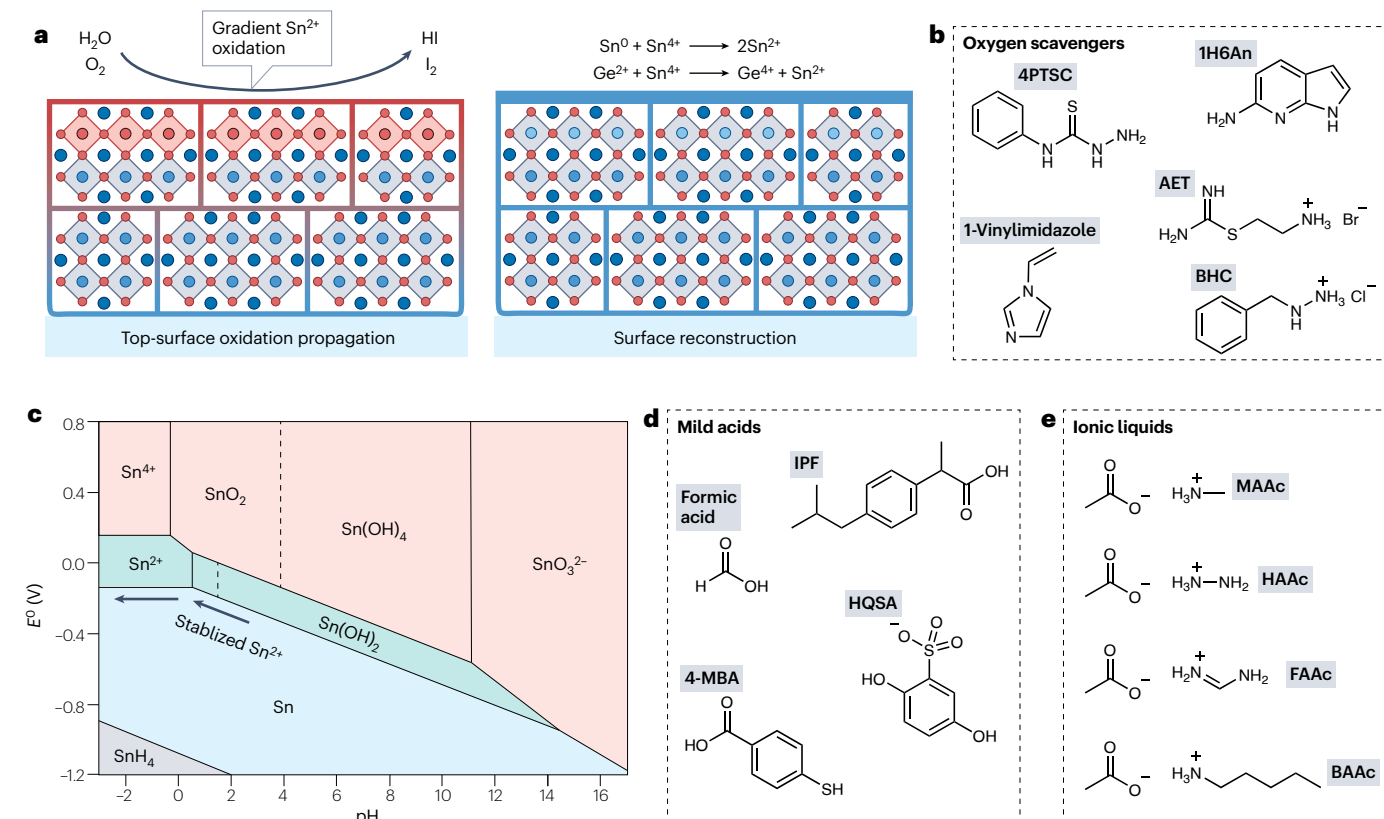


Fig. 4 | Antioxidation strategies. **a**, Gradient surface oxidation of a tin halide perovskite (THP) film with oxygen scavenger additives at grain boundaries and interfaces in ambient conditions. **b**, Representative oxygen scavengers to protect THPs. **c**, Pourbaix diagram of tin, showing Sn^{4+} (pink), Sn^{2+} (green) and Sn

(blue) regions. The standard electrode potential and pH data are extracted from ref. 37. **d**, Representative mild acid additives to stabilize Sn^{2+} species in the THP precursor solutions. **e**, Typical ionic liquid additives used to protect THP films from oxidation. E^0 , standard reduction potential.

Antioxidation approaches

Sn^{4+} impurities in precursor solutions can be readily eliminated by adding excessive metallic tin powder, but the formation of Sn^{4+} species in THP films remains inevitable owing to the spontaneous Sn^{2+} oxidation during crystallization and post-treatment²⁸. Extrinsic antioxidation can be achieved by incorporating oxygen scavengers in the precursor solution to persistently reduce Sn^{4+} for a limited period, whereas an acidic environment can increase the redox potential of $\text{Sn}^{4+}/\text{Sn}^{2+}$ to suppress oxidation of THPs intrinsically for a longer term. However, the preferential accumulation of Sn^{4+} species at the top surface caused by gradient oxidation is routinely observed (Fig. 4a). Thus, surface construction strategies are required to remove surface Sn^{4+} impurities and form passivation or encapsulation interlayers, minimizing interfacial PV performance losses and enhancing the oxidation resistance of THP films under operational conditions.

Oxygen scavengers and acidic antioxidation. To suppress Sn^{2+} oxidation, one strategy is to incorporate reducing agents in the precursor solution, where these oxygen scavengers continuously reduce Sn^{4+} species during solution processes. For instance, the commercial deoxidant dimethyl ketoxime, featuring an electron-rich oxime group ($\text{C}=\text{NOH}$), can effectively consume permeated oxygen molecules and reduce Sn^{4+} phase impurities during the crystallization of $\text{CsSnI}_{2.6}\text{Br}_{0.4}$ films⁷⁵. Compared with volatile reducing agents, introducing non-volatile

antioxidants in the precursor solution enables residual reducing agents at grain boundaries and on the film surface, forming in situ encapsulation barriers that persistently protect THP films from oxidation⁴⁸. Specifically, adding NaBH_4 in the precursor solution of FASnI_3 offers holistic protection during fabrication and ageing processes, enabling operationally stable THPSCs that retain more than 96% of the initial PCEs after 1,300 h under MPPT in nitrogen. Similarly, small-molecule antioxidants with strong reducing functional groups, such as hydroxybenzene, hydrazine and benzyl hydrazine, have been employed to suppress Sn^{4+} formation in the precursor solution and protect the resultant THP films^{76–78} (Fig. 4b). However, these sacrificial agents typically maintain low Sn^{4+} densities over limited ageing periods, whereas prolonged antioxidation requires more intrinsic regulation of the $\text{Sn}^{4+}/\text{Sn}^{2+}$ redox chemistry.

The redox activity of $\text{Sn}^{4+}/\text{Sn}^{2+}$ can be evaluated by its standard reduction potential (E^0), which is -0.93 V and 0.151 V in alkali and acidic media, respectively⁷⁹ (Fig. 4c). The negative E^0 value indicates spontaneous Sn^{2+} oxidation in alkali medium, whereas a positive E^0 value under acidic conditions refers to thermodynamically stable Sn^{2+} . Given a positive oxidation potential of $\text{Sn}^0/\text{Sn}^{2+}$ (0.13 V in acidic medium), the comproportionating reaction between Sn^{4+} and Sn is thermodynamically favourable, with an overall E^0 of 0.281 V. This electrochemical basis explains why adding metallic tin powder in the precursor solution stabilizes Sn^{2+} species. Whereas mild acids (pH 2–3) such as citric

acid, formic acid, acetic acid, salicylic acid and oxalic acid facilitate the reduction of Sn^{4+} , strong acids ($\text{pH} < 1$) generate excessive protons that disrupt colloidal stabilization in the precursor solution and impede the nucleation of THPs. Consistently, carboxylic and thiocarboxylic additives with moderate acidity, such as 4-mercaptobenzoic acid and ibuprofen, have been developed to attain a delicate reducing environment, stabilizing Sn^{2+} in the precursor solution without perturbing the crystallization kinetics of THPs^{80,81} (Fig. 4d). Alternatively, ionic liquids bearing acetate and formic acid anions ensure a mild acidic condition without generating excess protons, whereas reducing cations such as hydrazine and butylammonium can be introduced at A sites of THPs to alleviate surface Sn^{2+} oxidation^{62,78} (Fig. 4e).

Surface reconstruction. To suppress non-radiative recombination at the interface and improve the durability of THP films, surface reconstruction strategies – such as reducing interlayers and chemical polishing – have been developed to eliminate Sn^{4+} species and passivating undercoordinated Sn^{2+} sites at the interface. For instance, 1 nm tin interlayers have been deposited at the buried interface and top surface of THP films using thermal evaporation^{82,83} and plasma deposition⁸³. These conformally deposited tin interlayers reduce Sn^{4+} sites at both interfaces, forming bilateral oxidation barriers to protect the wrapped THP film. Moreover, alloying Ge^{2+} with Sn^{2+} can reduce Sn^{4+} species during the crystallization of CsSnI_3 via spontaneous $\text{Ge}^{2+}/\text{Ge}^{4+}$ redox reactions, forming an in situ encapsulation of the GeO_x interlayer (<5 nm) against the external oxidative environment^{84,85} (Fig. 4a).

Furthermore, surface post-treatment of Lewis base, reducing passivation agents can neutralize surface undercoordinated Sn^{2+} (suppressing non-radiative recombination), while also eliminating Sn^{4+} impurities and immobilizing iodide migration at the interface. This post-treatment therefore improves the operational stability of THP films under light and thermal stresses^{13,76}. Given the potential corrosion from post-treatment solvents, intact Sn^{4+} surface de-doping can also be achieved by physically deposited interlayers. For instance, a thin FACl capping layer (~3 nm) deposited by thermal evaporation can selectively react with surface SnI_4 species to form volatile $\text{SnI}_4 \cdot x\text{FACl}$ complexes, which can be easily removed under mild annealing⁸⁶. This chemo-thermal approach enables gentle surface polishing that mitigates Sn^{4+} impurities and passivates dangling bonds at the top surface, thus enhancing the oxidation resistance of THP films.

2D–3D perovskite heterostructures

Compared with their 3D counterparts, 2D THPs are intrinsically more resistant to oxidation owing to their lower disproportionation enthalpies and the spatially inhibited oxygen permeation by bulky organic spacer cations⁸⁷. Benefiting from enhanced coordination between bulky organic spacer cations and $[\text{SnX}_6]^{4-}$ octahedral units, these 2D phases also exhibited suppressed trap states associated with V_{Sn} , owing to their relatively lower defect formation energies (ΔH_f). For instance, PEA_2SnI_4 (PEA = phenethylammonium) possesses a lower ΔH_f of V_{Sn} value (0.46 eV) under tin-rich conditions, compared with that (0.26 eV) of MASnI_3 . This lower ΔH_f value leads to a reduced V_{Sn} concentration of 10^{14} cm^{-3} in PEA_2SnI_4 , which is three orders of magnitude lower than that of MASnI_3 (10^{17} cm^{-3}) (ref. 88). In this context, constructing bulk 2D–3D THP heterostructures, where 2D phases are embedded into a 3D THP matrix, enhances the oxidation stability of THP layer, while also providing high-quality crystalline templates to guide the oriented growth of THP films with suppressed trap states.

Templated growth and spacer engineering. Owing to high octahedral lattice match, the deployment of pre-nucleated 2D perovskites can induce the quasi-epitaxial growth of 3D perovskites, leading to highly crystalline THP films with low defect densities⁸⁹. For instance, surface treatment of wet perovskite films using alkylammonium cations, such as propylammonium and pentafluoro-phenoxethylammonium, can induce a preferential nucleation orientation of FASnI_3 along the (100) plane, underscoring the template effect of 2D intermediate phases^{10,11}. Beyond surface-controlled growth that forms bilayer 2D–3D PHSs, this template effect can be further amplified by directly adding 2D spacer cations in precursor solutions, inducing the formation of bulk 2D–3D PHSs with substantially reduced defect densities and improved microstructure⁹⁰ (Fig. 5a). The incorporation of pseudohalide anions, such as SCN^- , COOH^- and BF_4^- , can lower the cleavage energies of quasi-2D perovskites and enable the early nucleation of 2D templates at the liquid–air interface, thus guiding the orientated growth of 3D phases from the top surface towards the substrate^{91,92}. Moreover, the out-of-plane orientation of 3D phases can be promoted by designing volatile intermediate 2D phases through introducing Cl^- and Ac^- anions in the 2D phase precursors, facilitating Ostwald ripening during subsequent annealing at 100 °C and thus inducing the controlled growth of uniform and large-grain THP films⁹³.

Despite improved film quality of 2D–3D PHSs, conventional 2D THPs suffer from relatively low charge-carrier mobility owing to quantum confinement induced by high exciton binding energies at their spacer interlayers⁵⁹. Conjugated spacer cations have been developed to address this issue by improving the charge transport between 2D and 3D domains, aiming to suppress non-radiative recombination within 2D–3D PHSs (Fig. 5b). Compared with the conventional PEA^+ spacer cation, thiophenethylammonium (TEA^+), which features a sulfur-containing heterocycle, exhibits a larger dipole moment and reduced steric hindrance, enabling promoted interlayer conjugation and thus improved out-of-plane charge transfer of 2D phases^{46,94}. By decorating electron-withdrawing fluorine on the benzene ring of PEA^+ , the dipolar FPEA^+ spacer can accelerate nucleation kinetics of 2D phases and enrich 2D segments at the top surface region, leading to the formation of heterogeneous 2D–3D PHSs with improved interfacial charge transport and oxidation resistance⁹⁵. Consistently, in TEA -based 2D–3D THP systems, the incorporated FPEA^+ cations were expelled to the top surface during crystallization, forming interfacial dipoles that optimize band alignment between THP and ICBA layers and thus delivering an increased V_{oc} of 0.97 V and a certified PCE of 15.7%¹².

Although monoammonium spacer cations form Ruddlesden–Popper phase 2D perovskites, diammonium spacer cations such as ethylenediammonium (EDA^{2+}) and 1,4-butanediammonium have been employed to synthesize Dion–Jacobson phase 2D perovskites with shortened spacer interlayers^{96,97}. Despite improved out-of-plane charge-carrier mobility, these Dion–Jacobson phase 2D THPs have not demonstrated a pronounced template effect on the crystallization of FASnI_3 , suggesting the potential lattice mismatch between Dion–Jacobson phase 2D and 3D THPs. Meanwhile, mixed-spacer systems have been explored to promote uniform and ordered nucleation of 2D THPs with narrower n -value distributions, where n refers to the octahedral layer number between spacer cations^{98–100}. Despite improved device durability, these mixed-spacer 2D–3D PHSs generally delivered inferior PV performance compared with their single-spacer counterparts, owing to increased lattice mismatch and strain accumulation at the 2D–3D interfaces¹⁰¹.

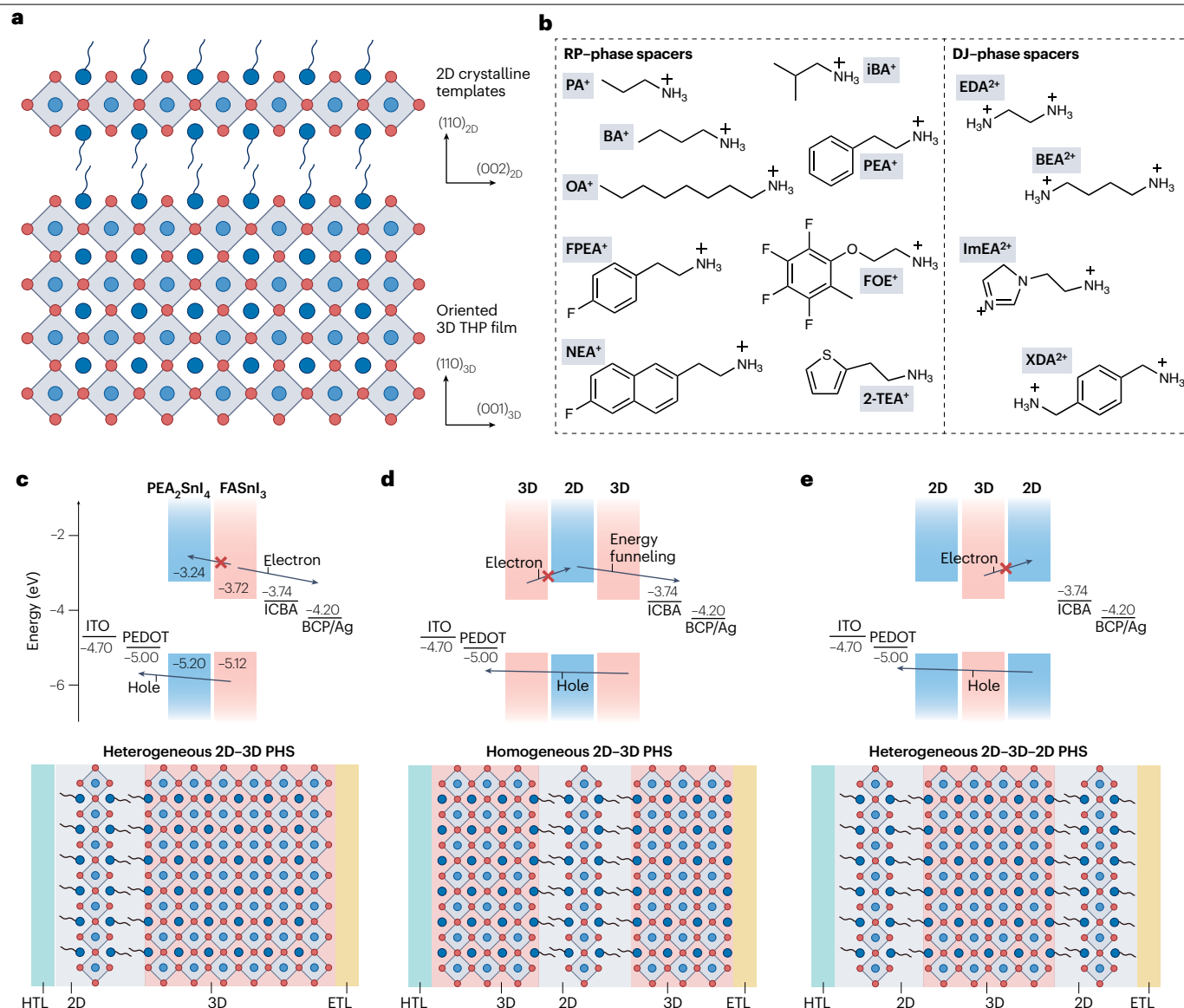


Fig. 5 | Design of 2D–3D tin halide perovskite heterostructures. **a**, Pre-nucleated 2D phases serve as crystalline templates to induce the controlled growth of a 3D tin halide perovskite (THP) matrix with preferential orientation along the (110) plane, leading to the formation of bilayer or bulk 2D–3D perovskite heterostructures (PHSs). **b**, Representative spacer cations of RP and DJ-phase 2D THPs. **c–e**, Crystal structures of heterogeneous 2D–3D (panel **c**), homogeneous

2D–3D (panel **d**) and heterogeneous 2D–3D–2D (panel **e**) THP heterostructures, and their energy band alignment with charge transport layers. EDA²⁺, ethylenediammonium; ETL, electron transport layer; HTL, hole transport layer; PEA⁺, phenethylammonium; PEDOT, poly(3,4-ethylenedioxythiophene); TEA⁺, thiophenethylammonium.

Phase purity and distribution. The n -value distribution of 2D phases and their spatial locations are profound in governing charge-carrier dynamics of bulk 2D–3D PHSs. Considering the insulating nature of low- n 2D phases ($n \leq 2$), the deployment of high- n 2D phases ($n \geq 3$) with higher mobility is essential to facilitate efficient charge transport in the bulk THP films. In this regard, anion engineering by incorporating pseudohalides is a promising way to lower the formation energies of high- n 2D THPs, enabling more conductive quasi-2D–3D PHSs for efficient THPSCs^{71,92}. However, precise control over the phase purity of 2D THPs remains challenging, whereas a broad n -value

distribution markedly increases energy disorder, impairing charge transfer between 2D and 3D domains. In this context, phase-pure 2D–3D PHSs can be deterministically fabricated by dispersing pre-synthesized 2D perovskite seeds in the precursor solution before deposition, where non-coordinative low-DN solvents, such as acetonitrile (DN of 14.1 kcal mol⁻¹), γ -butyrolactone (DN of 17.8 kcal mol⁻¹) and 2-methoxyethanol (DN of 19.7 kcal mol⁻¹), can be employed to preserve the native chemical coordination of these 2D seeds¹⁰².

From the crystallization control perspective, a vertically uniform distribution of 2D nuclei is critical to ensure a holistic template effect

during the top-down crystallization of the 3D phase matrix, facilitating the formation of dense and pinhole-free THP films. Due to their sluggish nucleation kinetics, PEA-based 2D phases tend to accumulate at the buried interface during crystallization, rendering an insufficient template effect at the liquid–air interface and, thus, abundant defects at the top region of THP films. This 2D and 3D phase segregation can be mitigated by incorporating Cs^+ cations to promote the coagulation of 2D and 3D colloids in the precursor solution. The resulting enlarged mixed 2D–3D colloidal clusters synchronize the nucleation kinetics of 2D and 3D phases, therefore amplifying the template effect of 2D phases to regulate the growth of 3D phases¹⁶. Alternatively, the nucleation of 3D phases can be deliberately slowed by introducing functional additives that strongly coordinate with FA^+ cations while interacting weakly with 2D spacer cations (PEA^+ or TEA^+)⁴⁶. This approach ensures the preferential formation of 2D templates that subsequently induce epitaxial growth of the 3D phase, leading to the formation of homogeneous 2D–3D PHSS with reduced trap states and improved oxidation resistance.

The valence band and conduction band of 2D perovskites are primarily determined by central and edge SnI_6 octahedral units, respectively¹⁰³. Compared with their 3D analogues, 2D THPs exhibit a slight downshift in the valence band but a pronounced upshift in the conduction band, owing to strong interaction between the spacer cations and edge octahedral domains⁵⁹. From energy band alignment perspectives, the presence of 2D phases at the buried interface does not hinder hole extraction⁹⁵. In contrast, introducing a 2D THP capping layer at the top surface can impede electron transfer owing to the large conduction band offset, for instance, a conduction band variance of 0.48 eV between PEA_2SnI_4 and FASnI_3 (ref. 16) (Fig. 5c). Thus, heterogeneous 2D–3D PHSS with a gradient distribution of 2D segments near the bottom interface can still maintain efficient interfacial charge transport, whereas top surface accumulation of 2D segments is energetically unfavourable. In homogeneous 2D–3D PHSS, embedded 2D segments partially impair charge-carrier mobility but allow energy funnelling from the wide-bandgap 2D phases to narrow-bandgap 3D domains via Förster resonance energy transfer¹⁶. The efficient charge-carrier dissociation and transport are evidenced by their long charge-carrier diffusion lengths beyond 450 nm. For heterogeneous 2D–3D–2D PHSS, electron transfer at the perovskite–ETL interface can be partially inhibited to reduce photocurrent output, although in situ grown 2D phase capping layers can enhance the resistance of THP films against external stimuli¹⁰⁴.

Interfacial engineering

Interfacial PV performance losses of THPSCs primarily originate from surface trap-assisted non-radiative recombination and suboptimal band alignment between the perovskite absorber and the adjacent transport layers⁴⁴. Various surface passivation strategies have been developed to mitigate trap states at both the top surface and the buried interface. Furthermore, designing suitable ETLs and HTLs enables minimal energy band offsets and promotes efficient charge transport at both interfaces of THP films (Fig. 6a). In an inverted p–i–n device configuration, compact and chemically robust ETLs can effectively protect the underlying THP films from extrinsic degradation. In parallel, tailoring the surface chemistry of HTLs can regulate the crystallization kinetics of THP films and reinforce buried interface interactions to enhance structural integrity and durability of the stacking layers against external stimuli¹⁷.

Top surface engineering. Due to its high chemical activity, the top surface of THP film can be partially corroded by the commonly used

post-treatment solvent, isopropanol. To protect the vulnerable THP films, non-polar alternatives such as hexafluoro-2-propanol with reduced nucleophilicity have been developed to dissolve alkylammonium halides and passivate top surface trap states¹⁰⁵. To achieve synergistic defect passivation, a sequential passivation protocol using acetylacetone and EDA solvents has also been designed¹⁰⁶. Specifically, the initial acetylacetone treatment reduces the surface trap density and enlarges the grain size to enhance photocurrent, followed by EDA treatment to passivate undercoordinated Sn^{2+} sites and regulate surface potential, thereby suppressing interfacial photovoltage losses. To ensure sufficient solubility of passivation agents, mixed solvents of 5% 2-methyl-2-butanol and 95% chlorobenzene have been employed to dissolve iso-butylammonium iodide (iso-BAI). Such a mixed solvent system triggered a dissolution–recrystallization process for top surface passivation, in which 2-methyl-2-butanol selectively dissolves Sn^{4+} species and facilitates the subsequent surface recrystallization by cation exchange between iso-BA⁺ and FA^+ (ref. 42).

To avoid solvent-induced corrosion of crystalline THP films, passivation agents can also be incorporated in the anti-solvent, triggering surface recrystallization during the dynamic spin-coating process. For instance, tetraethylammonium (TEA^+) cations have been introduced in a mixed anti-solvent of *N,N*-diethylformamide and isopropanol, effectively suppressing Sn^{2+} oxidation and forming gradient 2D–3D PHSS to facilitate electron transfer and suppress ion migration at the interface, yielding a high V_{oc} of 1.04 V and thus an improved PCE of 15.48% with negligible hysteresis⁴³. To alleviate self p-type doping induced by Sn^{4+} species, gradient THP heterostructures with electron acceptors have been designed to compensate self p-doping and induce a more n-type quasi-Fermi level splitting at the top surface of THP film, thus establishing a cascade band alignment to enhance electron extraction efficiency. Typically, pyridyl-substituted fulleropyrrolidines and non-fullerene acceptor Y6 have been incorporated in the precursor solution or anti-solvent to regulate crystallization and optimize interfacial band alignment, effectively suppressing interfacial non-radiative recombination losses and thus enabling high PCEs exceeding 16%^{41,107}.

In addition to the commonly used ICBA, functionalized fullerenes have been developed as coordinative ETLs to enhance their interfacial interactions with the underlying THP film, suppressing surface trap states and optimizing energy band alignment (Supplementary Fig. 1a). For instance, fullerene-*n*-butyl-pyridine (C_{60} -BPY) can strongly anchor on the THP surface through chemical coordination between its pyridine moiety and Sn^{2+} sites¹⁰⁸. Owing to shallower lowest unoccupied molecular orbital level, employing C_{60} -BPY and C_{60} as bilayer ETLs resulted in improved energy band alignment at the top interface, and thus increased V_{oc} by 70 mV in the target device¹⁰⁸. Meanwhile, the volatility and coordination of the employed solvents affect the packing orientation and film morphology of fullerene-based ETLs. Compared with toluene and chlorobenzene, anisole with an electron-donating methoxy group can form stronger interaction with fullerene cages, effectively regulating the orientations of ICBA-based ETLs and yielding higher electron mobilities¹⁰⁹.

In contrast to spherical fullerene-based ETLs, non-fullerene acceptors with planar molecule structures and multiple electron-withdrawing functional groups can form continuous and conformal capping layers on THP films, facilitating electron transfer at the THP–ETL interface. For instance, non-fullerene triple-acceptor polymers, featuring thiophene and fluorine functional groups, exhibited enhanced electron mobilities of $3.17 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ that are three orders of magnitude higher than that of ICBA ($6.9 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$). Benefiting from reduced band

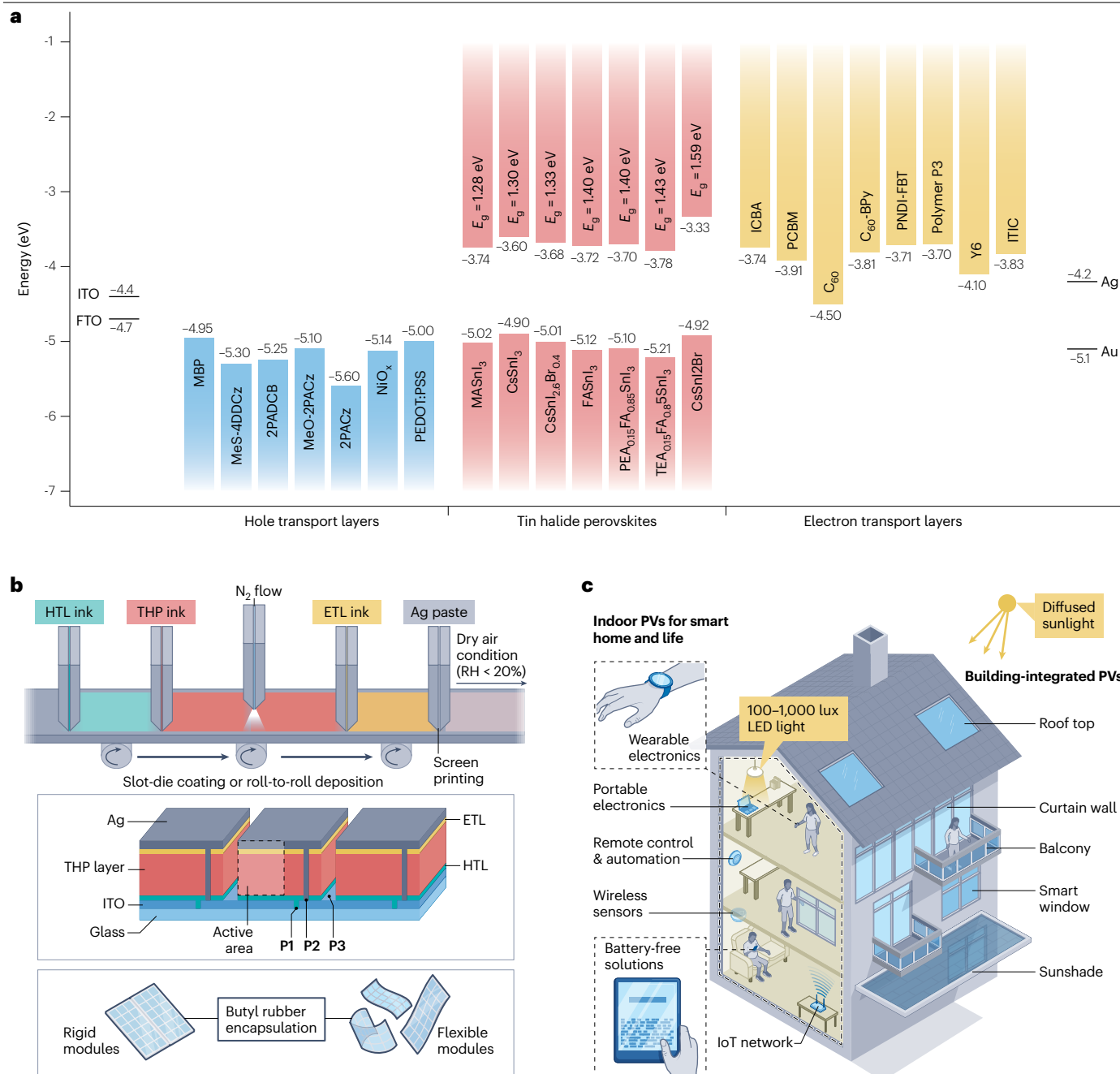


Fig. 6 | Energy band alignment, scalable fabrication and future applications. **a**, Energy levels of tin halide perovskites (THPs) with varying chemical compositions (red), and the highest occupied molecular orbitals of hole transport layers (HTLs; blue) and lowest unoccupied molecular orbitals of electron transport layers (ETLs; yellow). Data available in Supplementary Table 4. **b**, Schemes of scalable printing and patterning processes of tin halide

perovskite solar cells (THPSCs). **c**, Promising niche applications of THPSCs. C₆₀-BPy, fullerene-*n*-butyl-pyridine; ICBA, indene-C₆₀ bis-adduct; IoT, Internet of Things; NiO_x, nickel oxide; PEA, phenethylammonium; PEDOT:PSS, poly(3,4-ethylenedioxythiophene) polystyrene sulfonate; TEA, thiophenethylammonium.

offset and improved interfacial coupling with THP film, THPSCs based on these non-fullerene ETLs delivered high PCEs above 16%¹¹⁰. Nevertheless, an n-type polymer-based ETL of PNDI-FBT exhibited a much lower synthesis cost that is only 6% that of ICBA, along with enhanced electron mobility and intrinsically photothermal stability, rendering it

a potentially more cost-effective ETL for THPSCs⁵³. Whereas the functionalization of fullerene-based ETLs requires multistep purification and separation processes, non-fullerene acceptors provide greater structural tunability to optimize electronic properties, interfacial compatibility and chemical stability, offering lower-cost ETL alternatives

for efficient and durable THPSCs⁵³. Considering the enhanced covalent interactions of planar non-fullerene acceptors compared with spherical fullerene derivatives, these non-fullerene ETLs enable better top surface passivation¹¹⁰ and thus improved resistance of the underlying THP films against oxygen and moisture.

Buried interface modification. Inserting an interlayer that can chemically bridge the underneath HTL and the overlying THP light absorber can passivate trap states at the buried interface. For instance, aromatic diammonium cations such as 4-(aminomethyl)pyridinium can simultaneously interact with the PSS[−] chain of PEDOT:PSS and halide anions of THPs by ionic bonding, suppressing defect formation and facilitating hole extraction at the buried interface⁵² (Supplementary Fig. 1b). Meanwhile, post-treating PEDOT:PSS with carboxylic salts that contain formate, acetate or citrate acidic anions can introduce uniformly distributed metal chelation sites on the substrate, facilitating fast nucleation and subsequent bottom-up growth of THPs to strengthen interfacial coupling^{111,112}. In addition, the introduction of PEA⁺ cations on the HTL before perovskite deposition can induce the preformation of 2D perovskite interlayers, templating the growth of 3D phases from the buried interface⁶⁷. However, these 2D interlayers can be partially dissolved by perovskite precursor solutions, rendering inferior thickness control. In this scenario, 2D spacer cations that are more resistant to polar solvents should be selected to achieve robust 2D THP interlayers, inducing epitaxial growth of high-quality THP crystals from the bottom to the top surface.

Despite its intrinsic hygroscopicity¹¹³, PEDOT:PSS remains the most widely used HTL of THPSCs. To suppress interfacial degradation and optimize interfacial band alignment, carbazole-based SAMs such as MeO-2PACz and 2PACz have been employed as bridging interlayers that form strong chemical interactions with both PEDOT:PSS and THP layers¹¹⁴ (Supplementary Fig. 1c). Specifically, the anchoring PO₃[−] group of these SAMs tends to coordinate with the S⁺ terminals of PEDOT:PSS, whereas the methoxy (CH₃O[−]) terminals of MeO-2PACz interact with undercoordinated Sn²⁺ at the bottom surface of THP film, thus enhancing chemical coupling at the buried interface. Similarly, an amino acid-derived molecule, S-benzyl-L-cysteine, can serve as a bilateral coordinating interlayer, interacting with both PEDOT:PSS and THPs through its carboxyl and amine functional groups¹¹⁵. In addition, a methylthio-decorated carbazole SAM has been inserted beneath the PEDOT:PSS HTL, strengthening its interfacial contact with the FTO glass substrate and simultaneously improving energy band alignment to facilitate hole extraction¹¹⁶.

Inorganic NiO_x represents a robust alternative to organic PEDOT:PSS. However, direct contact between NiO_x and THP layers can induce gradual Sn²⁺ oxidation at the interface, accelerated by the diffusion of reactive oxygen and hydroxide species¹⁵. To mitigate interfacial degradation, a cyanovinyl phosphonic acid-based SAM of (E)-(2-(4',5'-bis(4-(bis(4-methoxyphenyl)amino)phenyl)-[2,2'-bithiophen]-5-yl)-1-cyanovinyl)phosphonic acid (MBP) has been introduced as a bridging interlayer to suppress oxidation from the underlying NiO_x HTL¹⁷. Besides improved band alignment, the homogeneous molecule packing of MBP on the NiO_x layer creates a superwetting underlayer that guides the uniform growth of high-quality THP films with reduced defect densities, leading to a high certified PCE of 17.71% as well as prolonged device lifespan with a T₉₄ of 1,550 h under operational conditions. Beyond molecular design, the wettability of SAM layers can be improved further by incorporating 6-phosphonohexanoic acid additives or pre-depositing Al₂O₃ nanoparticles (30–100 nm), both of which facilitate more ordered SAM packing on the substrate^{117,118}.

Notably, owing to the upshifted valence band of THPs, commercial SAMs used in lead-based perovskite solar cells generally require further chemical modification to raise their highest occupied molecular orbital levels²¹, narrowing band offset and thus promoting hole extraction at the SAM–THP interface towards efficient THPSCs.

Scalable fabrication and commercialization

Validating the scalable manufacturing of large-area THP modules using low-cost printing techniques is essential for commercialization. Initial efforts using blade and slot-die coating demonstrated the feasibility of depositing large-area THP films. However, the resulting interconnected mini-modules (25 cm²) exhibit a low PCE of 5.7%¹¹⁹. Mixed polar solvent systems have been introduced to regulate crystallization kinetics^{54,56}. Nevertheless, uncontrolled solvent evaporation during blade coating can lead to non-uniform nucleation, producing pinhole-rich and defective films⁵⁶. To broaden the processing window, a two-step blade coating strategy, involving sequential deposition of SnI₂ and FAI–PEAI in N₂, has achieved more reproducible PCEs exceeding 10%⁵⁵. Despite improved crystallization control, this approach prolongs the exposure of Sn²⁺ precursors to oxygen and moisture, thereby accelerating oxidation and inducing additional efficiency losses. Slot-die coating has also been demonstrated for large-area THPSCs, although device performance remains limited, with a PCE of only 2.31% for 5 cm² devices¹²⁰.

As the cyclic degradation of THPs is mediated by both oxygen and moisture under ambient conditions, controlling humidity during fabrication is critical to suppress oxidation propagation within THP films⁴⁰. Processing in dry air (relative humidity level < 20%) and incorporating oxygen scavenger additives can effectively retard degradation pathways, enabling scalable printing of THP modules on both rigid glass and flexible substrates. These modules can be further integrated with mild laser patterning and butyl rubber encapsulation, offering a viable route towards high-throughput manufacturing (Fig. 6b).

Benefiting from high efficiencies under low light intensities, being lightweight, compatibility with flexible substrates and fewer safety concerns compared with their lead-based counterparts, THP modules are well positioned to complement outdoor silicon PVs in niche applications, including indoor PVs, building-integrated PVs and seamless power supplies for wearable electronics, and Internet of Things (IoT) networks¹²¹ (Fig. 6c). Targeting these emerging applications allows THPSCs to accumulate manufacturing and deployment experience at manageable production scales, offering a pragmatic and timely commercialization pathway. This strategy enables early market entry prior to full technological maturation and avoids direct competition with lead-based perovskites and the well-established silicon PV industry¹²². With continued improvements in efficiency and stability, THPSCs could expand to more demanding applications, including multi-junction PVs such as Si–THP tandems and all-THP tandems, and specialized environments such as space and underwater PVs, where restricted oxygen exposure can mitigate the intrinsic instability of THPs¹²³.

Summary and future perspectives

Despite progress in PV performance metrics, the practical deployment of THPSCs remains constrained by coupled limitations in efficiency, stability and scalable manufacturing, largely rooted in abundant bulk and surface defects, surface-enriched Sn⁴⁺ species, insufficient NIR light harvesting and suboptimal transport layers. These fundamental challenges exacerbate severe non-radiative recombination and accelerate chemical degradation under light and thermal stresses, deteriorating the long-term operational stability of THPSCs.

In the short term, continued efficiency improvements in THPSCs will rely on suppressing defect formation in both the bulk THP film and at their interfaces. Concurrently, device architectures inherited from lead-based perovskite solar cells need to be further optimized by developing high-mobility HTLs and ETLs that are energetically aligned with the upshifted conduction band and valence band of THPs. In parallel, developing low-cost non-fullerene alternatives to the conventional fullerene-based ETLs, combined with robust interfacial interlayers, can further improve device performance, durability and cost-effectiveness. By implementing more rigid and conjugated 2D templates to regulate crystallization, better light management could be achieved by preparing thick 2D–3D THP layers without compromising film quality and conductivity, promoting NIR light harvesting and minimizing photocurrent losses.

In addition to efficiency breakthroughs, the long-term operational stability of THPSCs also requires marked improvement. Mitigating degradation pathways activated under continuous light illumination, electrical bias, thermal stress and moisture exposure remains unresolved. Beyond intrinsic crystal lattice stabilization, advanced interfacial engineering strategies such as bilateral interfacial passivation and chemically inert charge transport layers will help suppress ion migration and interfacial redox reactions. In parallel, reliable physical encapsulation protocols should be developed to effectively prevent external oxygen permeation. Establishing standardized ISOS stability protocols, such as MPPT under one-sun light, damp heat and diurnal cycling, remains crucial for assessing the practical lifespans of encapsulated THPSCs.

Lastly, technological translation from laboratory-scale prototypes to real-world products relies on the development of industry-compatible upscaling techniques that can reproducibly prepare high-quality THP films. Under dry-air conditions, introducing coordinative and acidic solvents such as ionic liquids to prepare THP inks is expected to intrinsically impede the facile oxidation and hydrolysis processes during scalable printing. Meanwhile, establishing a fundamental understanding of the correlation of film uniformity, defect density and properties of THP films during scalable deposition by advanced in situ and/or operando characterizations can help close the efficiency gap between small-area prototypes and module-size PV products.

Beyond indoor and build-integrated PV applications, established material processing and thin-film fabrication techniques could enable THPs to be a highly attractive low-toxicity platform for emerging optoelectronic applications, including near-infrared photodetectors and phototransistors, biomedical sensors, artificial synapses and sustainable power sources for flexible electronics.

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Author contributions

P.C. drafted the manuscript supervised by L.W. and Y.Y. P.C. and D.H. edited the manuscript. All authors revised and commented on the manuscript and approved the final version of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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